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25 April 1955
SUPERSEDING
NAVORD OS 3715
31 March 1945

MILITARY SPECIFICATION

DETONATOR MK 27 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading and preparation for delivery of Detonator Mk 27 Mod 0 and the methods of inspection and tests upon which acceptance will be based. The Detonator Mk 27 Mod 0 is stab initiated.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-T-339 -- Tetryl (trinitrophenylmethylnitramine).

STANDARDS

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

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DRAWINGS

BUREAU OF ORDNANCE

LD 166538	-- Test Set (Detonator) Mk 136 Mod 0,
LD 168793	-- Detonator Container Mk 3 Mod 0,
LD 168794	-- Detonator Container Mk 2 Mod 0,
LD 284563	-- Test Set Mk 170 Mod 1,
438241	-- Detonator Mk 27 Mod 0,
553491	-- Test Ball,
	and all documents listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	-- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.
OD 5823	-- Sensitivity Test of Primers and Detonators using Test Set Mk 135 Mod 0 (Primer) and Test Set Mk 136 Mod 0 (Detonator).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	-- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

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3. REQUIREMENTS

3.1 General.

3.1.1 Detonators Mk 27 Mod 0 covered by this specification shall be manufactured in accordance with Bureau of Ordnance Drawing No. 438241 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Handling. -- Loaded detonators shall be packed by lot, marked for lot identification, and stored in desiccators (or in hermetically sealed dry air) until such time as they are loaded into fuzes or prepared for shipment.

3.3 Priming Mixture. -- The moisture content of the priming mixture at the time of loading shall be less than 0.3 percent as determined by the procedure of 4.5.1. The composition of the priming mixture shall meet the drawing requirements when subjected to the chemical analysis described in 4.5.3.

3.4 Lead Azide. -- The moisture content of the lead azide at the time of loading shall be less than 0.30 percent as determined by the procedure of 4.5.1.

3.5 Tetryl. -- The volatiles and moisture content of the tetryl at the time of loading shall not exceed 0.10 percent when tested as specified in 4.5.2.

3.6 Detonator Sensitivity. -- The sensitivity of the detonators shall be ascertained as prescribed in 4.5.4. The sensitivity value shall be used for data card record and process control. Deviations in sensitivity values shall not directly cause the rejection of detonators, but may be reason for checking materials, processes, and priming mixture composition for compliance with drawing and specification requirements.

3.7 Detonator Functioning. -- The detonator shall function on firing pin penetration when tested in Test Set (Detonator) Mk 136 Mod 0 using a two ounce test ball and a drop height of five inches. Testing and acceptance shall be in accordance with 4.5.5.

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3.8 Detonator Output. -- Each explosion of the detonator in the Test Set Mk 136 Mod 0 shall produce a perforation in the lead disc greater than 0.300 inch in diameter. The perforation diameter shall be the diameter of a circle of equal area as determined by use of the Test Set Mk 170 Mod 1. Testing and acceptance shall be in accordance with 4.5.6.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Military Standard MIL-STD-105, "Sampling Procedures and Tables for Inspection by Attributes".

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot," if a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction Sample. -- Before entering into quantity production a preproduction sample of one thousand (1000) Detonators Mk 27 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These detonators shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction Sample for a Subsequent Contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the

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manufacture of satisfactory material as covered by this specification.

4.3 Production Lot. -- For procurement purposes each lot of loaded detonators shall consist of 10,000 units plus those required for all test purposes. The production lot should not be confused with the lot for inspection purposes as defined in Military Standard MIL-STD-105.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedure for Government Inspection and Contractors Inspection shall be in accordance with Inspection by Contractor as contained in OP 400.

4.5 Tests.

4.5.1 Moisture Content Test for Priming Mixture and Lead Azide. -- Conduct at least one test for each lot of detonators and such additional tests as the Inspector considers warranted by variations in the process and materials. Collect an approximate two gram sample of the material being loaded in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the detonator. Weigh the container; open and expose the sample for drying at 50 ± 2 degrees C and at least 28 inches (Hg) vacuum until constant weight is reached. Cool the container in a desiccator; stopper and weigh. All weighings shall be accurate to ± 0.5 milligram. The presence of moisture in excess of 0.30 percent for the priming mixture or lead azide shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Detonators which have been loaded with the defective charges shall be rejected.

4.5.2 Volatiles and Moisture Content Test for Tetryl. -- Conduct at least one test for each lot of detonators and such additional tests as the Inspector considers warranted by variations in the process and materials. Test procedure shall be in accordance with that given in Military Specification JAN-T-339. The presence of volatiles and moisture in excess of 0.10 percent shall cause suspension of the charge loading until permanent corrective measures

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satisfactory to the Inspector have been completed. Detonators loaded with the defective charges shall be rejected.

4.5.3 Chemical Analysis of Priming Mixture. -- Chemical analysis of the priming mixture to determine conformity with composition requirements of the drawing shall be made at the request of the Inspector. The test will normally be employed to determine cause of marked deviations in sensitivity or functioning resulting from composition or workmanship deficiencies. Priming mixture batches and detonators loaded therefrom shall be rejected when the composition does not meet drawing requirements. Analytical procedures for determining the percentage of each constituent of the priming mixture shall be in accordance with 4.5.3.1 through 4.5.3.4.

4.5.3.1 Potassium Chlorate. -- Weigh accurately approximately one gram of the primer mixture which has been dried in accordance with 4.5.1. Transfer the sample to a tared ignited filtering crucible of approximately 15 milliliter (ml) capacity. Add 3 ml of a saturated solution of lead azide in distilled water prepared at a temperature of 25 ± 2 degrees C. Agitate for 60 seconds. If necessary break up any lumps with a glass rod, taking care not to disturb the asbestos mat. Apply suction. Repeat this procedure five times, making a total of six 3 ml extractions. Rinse the sample in the filtering crucible with a few ml of alcohol, then ether; dry for approximately 15 minutes at 95 degrees C; cool in a desiccator, and weigh. The percentage loss in weight, based on the weight of the original sample, shall be considered the percentage of potassium chlorate.

4.5.3.2 Lead Azide. -- Extract the residue from the potassium chlorate determination with 5 ml portions of saturated ammonium acetate solution at about 25 degrees C. Hot saturated ammonium acetate solution must not be used as it dissolves the antimony sulphide to some extent. Agitate each portion for about 30 seconds and repeat extractions until the filtrate no longer gives a yellow precipitate with potassium dichromate solution. The use of more than 60 ml of ammonium acetate should be avoided. Wash the residue with distilled water; rinse with a few ml of alcohol, then ether; dry for approximately 15 minutes at 100 degrees C; cool in a desiccator, and weigh. The percentage loss of weight, based on the weight of the original sample shall be considered the percentage of lead azide.

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4.5.3.3 Antimony Sulphide. -- Extract the residue from the lead azide determination with hydrochloric acid until free from antimony sulphide. Cold concentrated hydrochloric acid is used for the extraction. After neutralizing, ammonium sulfide may be used to detect antimony in the filtrate. Wash the crucible contents with hot concentrated hydrochloric acid. Ignite until free from sulphur, cool in a desiccator, and weigh. The percentage, based on the weight of the original sample, shall be considered the percentage of antimony sulphide.

4.5.3.4 Carborundum. -- Subtract the original weight of the ignited filtering crucible from the weight of the crucible and residue after the antimony sulphide determination. This difference, calculated as a percentage of the original sample, shall be considered the percentage of carborundum.

4.5.4 Detonator Sensitivity Test. -- Select fifty (50) detonators from each lot (no lot shall exceed 20,000 detonators) for the Bruceton (Staircase) Sensitivity Test using 1.0 inch increments. Use Test Set (Detonator) Mk 136 Mod 0, Bureau of Ordnance List of Drawings LD No. 166538, with Test Ball, Bureau of Ordnance Drawing No. 553491, piece 3. Test set operation and test procedures are described in OD 5823. Record, on the Ammunition Data Card specified in 5.4, the 50 percent firing height ($\bar{h}$) and the standard deviation (s) calculated from the test data.

4.5.5 Detonator Functioning Test. -- Samples shall be selected and tested from each lot in accordance with the Multiple Sampling Plan, Inspection Level II, AQL 0.25 percent defective as given in Military Standard MIL-STD-105. The initiation of the detonator shall be accomplished by the use of Test Set Mk 136 Mod 0 equipped with the test ball specified in 4.5.4. The two ounce ball shall be dropped from a height of five inches to initiate the detonator. Detonators which do not explode on the first penetration of the firing pin shall be classed as defective. Acceptance of the lot shall be in accordance with the sampling plan given above. The lead discs shall be retained and used to fill the sample requirements specified in 4.5.6.

4.5.6 Detonator Output Test. -- Samples shall be selected and tested in accordance with the Single Sampling Plan, Inspection Level I, AQL 1.0 percent defective as

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given in Military Standard MIL-STD-105. The lead discs retained from the functioning test of 4.5.5 shall be used to fill sample requirements. Where necessary, continue the test firing of 4.5.5 to provide the required number of discs. If an excess of discs are available from 4.5.5 random selection of the sample shall be practiced. A detonator which produces a perforation with an equivalent diameter of less than 0.300 inch, as determined by measurement with Test Set Mk 170 Mod 1, shall be classed as defective. Lot acceptance shall be in accordance with the sampling plan given above.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- One hundred (100) Detonators Mk 27 Mod 0 shall be packaged in accordance with Bureau of Ordnance List of Drawings LD No. 168793.

5.2 Packing. -- Five thousand (5000) Detonators Mk 27 Mod 0, packaged as specified in 5.1 shall be packed in accordance with Bureau of Ordnance List of Drawings LD No. 168794.

5.3 Marking. -- In addition to any special markings required by the contract or order, all containers shall be marked in accordance with Military Standard MIL-STD-129, "Marking for Shipment and Storage", and Code of Federal Regulations 49 CFR 71-78, "Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles".

5.4 Data Cards. -- Data cards, five by eight inches, printed as shown below shall be filled out for each lot of detonators. Two of these cards shall accompany the shipping ticket, three shall be forwarded to the Bureau of Ordnance, and five shall be furnished the Inspector.

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AMMUNITION DATA CARD

Mark No. _____ Spec. No. _____
Dwg. No. _____ Date (Month & Year) _____
Loaded by _____ Req. No. _____
Contract No. _____ No. in lot _____
No. in box _____ $\bar{h}$ = _____
Lot No. _____ s = _____

MATERIAL	MANUFACTURER	LOT NO.	DATE (MONTH AND YEAR)
SEALING DISC			
CUP			
LEAD AZIDE			
ANTIMONY SULPHIDE			
POTASSIUM CHLORATE			
PRIMING MIXTURE			
TETRYL			
CARBORUNDUM			

Remarks: (These are to include any changes in the manufacturing processes prescribed by this specification which the Inspector feels might affect the uniformity of the lot).

DATA CARD

6. NOTES

6.1 Exposure of Explosive Materials. -- Exposure of explosive materials to the atmosphere during the loading operation should be so controlled as to minimize the absorption of moisture.

Patent Notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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